Fiscal Year:	FY 2016	**Task Last Updated:**	FY 09/06/2016
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Iron Overload and Oxidative Damage: Regulators of Bone Homeostasis in the Space Environment		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	sbloom@tamu.edu	**Fax:**	FY 979-862-1692
PI Organization Type:	UNIVERSITY	**Phone:**	979-845-2871
Organization Name:	Texas A&M University		
PI Address 1:	Department of Health & Kinesiology		
PI Address 2:	400 Harvey Mitchell Pkwy, Suite 300		
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	05/31/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
Contact Email:	kevin.y.sato@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/31/2018 per F. Hernandez/ARC (Ed., 7/31/17) NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 5/5/16)		
Key Personnel Changes/Previous PI:	No changes		
COI Name (Institution):	Ford, John (Texas Engineering Experiment Station) Hogan, Harry (Texas Engineering Experiment Station) Turner, Nancy (Texas AgriLife Research)		
Grant/Contract No.:	NNX13AL25G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Recent work in bone biology field has demonstrated that cellular damage due to molecules called reactive oxygen species (so-called oxidative damage) is a key mechanism contributing to aging-related bone loss, as well as to that occurring with the loss of estrogen after menopause. We know very little about whether the loss of gravity as experienced in the space environment modulates oxidative damage to bone. A newly recognized source of oxidative damage in bone is that due to iron overload, or an increase in body iron stores, as has now been demonstrated in both International Space Station (ISS) crew members. Data published this month indicate that sub-pathological levels of serum ferritin (a marker for iron stores) contribute to accelerated bone loss and to incident vertebral fractures in healthy middle-aged men and women. We have preliminary data indicating that similar moderate elevations in body iron stores, as induced by a high iron diet, can result in loss of cancellous bone and diminished bone formation activity in rodents. The experiments in this proposal will utilize iron overload as an experimental intervention to exacerbate oxidative damage during a period of simulated microgravity (by hindlimb unloading in male and female mice) to define the sex-specific mechanistic role of oxidative damage as a key regulator of bone mass and, ultimately, fracture risk as modulated by the gravitational unloading of the space environment. In a final specific aim, we will test whether superimposing acute and continuous (over 42 days) radiation exposure exacerbates oxidative damage and bone decrements during hindlimb unloading. These experiments are deliberately designed to enable a rapid transition to flight experiments requiring minimal intervention during flight aboard ISS, should these ground-based simulations yield provocative findings to justify flight experiments. Iron overload can be maintained by dietary intake, and limiting the acute radiation exposure in Experiment 2 to just before the unloading period allows for testing of the same hypotheses in a flight experiment. The experiments outlined in this proposal address how microgravity affects key regulatory mechanisms governing alterations in bone (from OCB-2 in the Space Biology Plan) and also one of the National Research Council (NRC) Decadal Survey's highest priority recommendations, to expand our understanding of sex differences in adaptation to the space environment, particularly in the area of bone and long-term radiation risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have evidence that oxidative damage caused by molecules that oxidize DNA or other tissue components can contribute to the loss of bone mass and integrity that occurs with aging and with the loss of estrogen in post-menopausal women. There are some unfortunate individuals who experience pathological increases in body iron stores (with diseases like hemochromatosis) who, we believe, also experience increased oxidative damage to multiple tissues due to an excess of free iron molecules. However, we now know that even moderate (sub-clinical) elevations in body iron stores in healthy middle-aged individuals can result in accelerated bone density loss. Only women in these groups, however, exhibit an increased risk (5-fold) of incident vertebral fractures. These experiments in skeletally mature mice will yield important and detailed information on the impact of these moderate elevations in body iron on evidence for oxidative damage and for accelerated bone loss, with the unique added treatment of reduced weight bearing (simulating prolonged bed rest in humans). Because this project will be examining both male and female mice, we will be able to define if there is any sex difference in the response of bone to iron overload, with and without coincident disuse. Our final specific aim will test the impact of increased iron stores in combination with disuse and exposure to chronic, very low dose gamma radiation, simulating space radiation. Radiotherapy patients on Earth experience a 2.6-fold increase in fracture risk compared to age-matched peers. A better understanding of the additional impact of radiation on bone integrity, with and without iron overload, may lead to better protection of bone health in radiotherapy patients, reducing fracture risk.
	Tasks accomplished --Year 3 August-October 2015: Because the previous year's experiments resulted in generally positive effects of increasing iron stores on bone in mice, as opposed to negative effects in rats in our previous work, we chose to add a small dose-response experiment in case we were simply not providing enough iron (in the diet) to replicate the response seen in rats. We tested 3 different levels of high-iron diets: 300, 900, and 1200 mg Fe/kg chow, n=7 young adult female mice per group, which were completed in early October. Their data will be compared with previously generated weightbearing mice consuming diets with "normal" Fe (45 mg/kg chow) and "high Fe" (650 mg/kg). With this group we also tested an additional 4 mice for tolerance of a treadmill running protocol for mice on hindlimb unloading (HU) to verify ability to run with suspension tail ring in place (but wrapped in tape) during time on treadmill (went well; reached a "go" decision on adding exercised groups to upcoming cohorts). Femurs shipped to Indiana U (consultant MR Allen) for micro-CT scans; MRA ships final uCT results on sex comparison studies. Piximus scans pre- and post-7 weeks on different Fe dose diets reveal no real dose-response relationship between total body BMD/BMC and dietary iron intake; this was confirmed by limited uCT scans of these samples. The decision was made to proceed with existing diet (650 mg Fe/kg). Mechanical testing of sex comparison study (male, female) completed for mid-shaft femur (3-point bending) and for axial compression of femoral neck. September - December 2015: M.S. student (D. Saucier) working with Co-I Dr. John Ford on Monte Carlo simulations for continuous radiation exposure (Year 3 project) to calibrate degree of cobalt wire activation required for very low dose rate requirement. [This was the subject of Mr. Saucier's MS thesis, successfully defended in November, 2015.] Several meetings were held with Director of Comparative Medicine Program, animal facility personnel, and Radiation Safety Officers on logistics for converting former x-ray room to dedicated "continuous radiation" (CIRR) room while controlling for any hot spots outside the room. Dr. Ford worked diligently to obtain Radiation Permit for transporting activated cobalt wires from A&M Nuclear Science Center to animal facility. November 2015 – May 2016: Investigator meeting held on November 30, 2015 (consultants Sara Zwart and Matt Allen attend by phone) to review dose response data and to plan forward for labor intensive CIRR studies. Deliberate pause taken in start of final animal experiments and in more expensive serum/tissue assays for sex comparison study completed previous summer due to lack of funds (incremental funding strategy by NASA in late 2014). Principal Investigator (PI) applied for one-year no-cost extension; granted in late April, 2016 (new end date now 7/21/17). Plans evolve for additional mouse groups to be exercised (with treadmill running) during CIRR; animal use protocol amendments generated. Dr. Ford gains approved Radiation Permit and early May starts setting up radiation field in CIRR room (transporting activated wires, erecting concrete shielding, performing dosimetry, verifying hot spots). GAs working diligently on analyses on collected tissues from sex comparison study: Progress to date on liver/serum assays: homogenized liver submitted for atomic absorption spectrometry for total iron content; 20-50% of samples completed for serum ferritin, serum hepcidin, liver catalase, liver mitochondrial SOD, and liver glutathione peroxidase (latter 3 are markers of oxidative damage).

Task Progress:	Progress of bone histomorphometry/IHC: Cortical bone histomorphometry (mid-tibia) completed (including bone formation rate); cancellous bone samples embedded and most are sectioned; labor-intensive histomorphometry analyses still in progress. May – July 2016: Running first two cohorts of female mice on final experiment of project: impact of continuous radiation exposure (CIRR) at very low dose rate (~0.8 mRem/hour) for 28 days in weightbearing and hindlimb unloaded mice, half of each group on high iron diets. Piximus scans done in vivo before and after CIRR. Success rate on completing all days of hindlimb unloading very high (only 2 out of 69 or <3% exit HU early, due to tail issues). Tissues collected and stored for later analysis. Interim Results: Validating iron status in our experimental groups: Atomic absorption spectroscopy assays confirmed a mild-moderate elevation (~=20%) in liver iron in mice from Year 2-3 cohort eating the high iron diet, with the most significant gain noted in male HU mice fed the high iron diet. Bone outcomes on sex comparison study: We have limited new bone data to report on the sex comparison beyond those reported in last year's annual report, since expensive assay kit purchases had to be delayed due to lack of funds and immunostaining of bone will be done only if we have other tissue evidence of oxidative damage in some of our experimental groups. One interesting finding was derived from measuring systemic markers of bone turnover from the sex comparison study: C-terminal telopeptides (CTX) of type 1 collagen (a bone resorption marker) and N-terminal propeptide of type 1 procollagen (P1NP; a bone formation marker) in mouse serum samples. Results indicate that CTX was 10% greater in hi FE females ($p<0.05$) with no change in hi FE males, while P1NP was 15% greater in FE males ($p<0.05$) but not females. This suggests a sex-specific response in osteoclasts (resorbing cells, increased activity with hi Fe only in females) and osteoblasts (bone forming cells, increased activity with hi Fe only in males) to elevated iron stores. However, cortical bone formation rate at the mid-tibia periosteal surface did not differ between males and females consuming diets with normal and elevated iron content. We also noted sex-specific differences in alterations of bone mechanical properties in response to elevated iron stores. Specifically, ultimate load and stiffness at mid-femur (by 3-point bending) were 17% and 25% higher, respectively, in male FE mice, but no changes were observed in female mice on the high iron diet ($p<0.01$). Ultimate load at the femoral neck (axial loading at the femoral head) was 34% higher in male FE mice, but no changes were observed for female mice on FE diet, nor for stiffness in either sex ($p<0.05$). These results are described in detail in an abstract submitted to the American Society of Bone and Mineral Research annual meeting (Sept 2016 in Atlanta, GA). Mechanical testing on mid-shaft femurs revealed that ultimate load (a key measure of bone strength) was reduced by 15 and 24% in HU males and females, respectively, but elevated iron stores result in stronger bone at this site within weightbearing status. These differences were statistically significant in males only. Inability to detect significant changes in the female mice mid-shaft ultimate load may be due to greater variability in ultimate load values for the female cage control-high iron (CC+Fe) group. Ultimate load at the femoral neck, a clinically relevant site, was reduced by 15 and 24% in HU males and females, respectively, while the high iron diet produced a gain, but only in cage control animals. Again, these differences reached statistical significance only in males. What's intriguing about these results: gains in bone density (BMD) and improved geometry from the high iron diet translate to improved mechanical strength in males but NOT in females. This implies that material mechanical properties might be mildly impaired in females on the high iron diet, countering the beneficial impact of higher BMD or more advantageous bone geometry. We note that our previous data in rats did not show significant negative impacts on bone mass until a high iron diet was combined with fractionated gamma radiation exposure (total dose 3 Gy). Hence we feel it is important to move ahead and assess the impact of combined continuous irradiation and elevated iron stores in hindlimb unloaded and weight-bearing (WB) mice. Results of dose-response studies (are we using the best "dose" of iron, added to mouse diet?): When we added in one lower iron dose group (300mg Fe/kg diet) as well as two higher doses (900 and 1200 mg Fe/kg) for eight weeks, we could demonstrate no consistent dose-response effect on total body BMC. Micro-CT studies on these animals' femurs confirmed this conclusion. Hence we made the decision to continue with the 650 mg Fe/kg diet we had been using, to minimize any negative side effects of higher iron intakes (although we observed no ill effects in these cohorts). Plans moving forward (Yr 4—no-cost extension year): Finishing liver/serum assays and cancellous bone histomorphometry on sex comparison study [WB and HU mice, on normal and high Fe diets]: Now that the final funding increment on this project has been received (early July 2016), we can move ahead with all planned analyses. By 11/1/2016, markers of oxidative stress in liver and serum will be completed by Dr. Turner's lab; we hope to finish bone histomorphometry work in the PI's laboratory by 12/20/16. Finishing final animal experiments [CIRR in WB and HU mice, on normal and high Fe diets]: Our final experiment (continuous exposure to very low dose rate gamma radiation [CIRR] over 28 days) will provide unique and valuable data. The final two cohorts of mice should be completed by December 15, 2016. Tissue analyses will consume the next 6 months, as performed by Turner (liver, serum assays), Hogan (mechanical testing), and Bloomfield (bone histomorphometry) laboratories. Corollary/tissue sharing projects: Given the cost and labor-intensive nature of constructing and maintaining this continuous radiation facility, we are working to maximize the data yield by actively sharing tissues with four other laboratories/ investigators and building on our own piggyback projects. Ms. Rihana Bokhari, the senior doctoral student on this project (and an NSBRI--National Space Biomedical Research Institute-- Space Life Science Doctoral Fellow), has been awarded a $5000 award from NASA via the American College of Sports Medicine for an add-on project testing the impact of CIRR on the bone response to exercise.
Bibliography Type:	Description: (Last Updated: 01/20/2026)

Abstracts for Journals and Proceedings	Bokhari RB, Metzger CE, Allen MR, Lenfest S, Hogan HA, Turner ND, Zwart S, Bloomfield SA. "Moderate elevations in iron stores improve skeletal integrity in mice even during disuse." Presented at the 37th Annual Meeting of the American Society for Bone and Mineral Research, Seattle, Washington, October 9-12, 2015. Journal of Bone and Mineral Research. 2015;30 (Suppl 1). , Oct-2015
Articles in Peer-reviewed Journals	Bloomfield SA, Martinez DA, Boudreaux RD, Mantri AV. "Microgravity stress: Bone and connective tissue." Compr Physiol. 2016 Mar 15;6(2):645-86. Review. http://dx.doi.org/10.1002/cphy.c130027 ; PubMed PMID: 27065165 , Mar-2016
Articles in Peer-reviewed Journals	Macias BR, Lima F, Swift JM, Shirazi-Fard Y, Greene ES, Allen MR, Fluckey J, Hogan HA, Braby L, Wang S, Bloomfield SA. "Simulating the Lunar environment: Partial weightbearing and high-LET radiation induce bone loss and increased sclerostin-positive osteocytes." Radiat Res. 2016 Aug 18. [Epub ahead of print] http://dx.doi.org/10.1667/RR13579.1 ; PubMed PMID: 27538114 , Aug-2016
Awards	Bloomfield SA. "American College of Sports Medicine Citation Award, May 2016." May-2016
Dissertations and Theses	Saucier D. "Monte Carlo irradiation model for long-term low dose environments for space life science applications." MS Thesis, Texas A&M University, College Station, TX, November, 2015. , Nov-2015